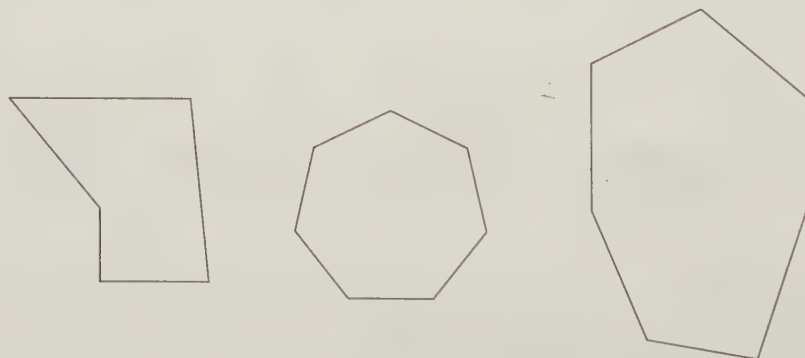




9

CHAPTER 9 Polygons



All of the names of the parts of a general polygon are the same as in the polygons we've learned about so far (triangles and quadrilaterals). The segments that form the boundary are **sides**, which meet at **vertices** to form the **interior angles**. If we connect two vertices that are not adjacent on the polygon, we form a **diagonal**, such as diagonal $\overline{AC}$ in Figure 9.2.

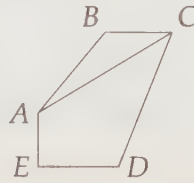


Figure 9.2: Polygon Diagonal

Polygons are classified by the number of sides they have. Here are some names given to different polygons.

Number of Sides	Polygon Name
3	triangle
4	quadrilateral
5	pentagon
6	hexagon
7	heptagon
8	octagon
9	nonagon
10	decagon
12	dodecagon
n	n -gon

A polygon in which all the sides are equal and all the angles are equal is called a **regular polygon**. We've already seen two examples of regular polygons: equilateral triangles and squares. Figure 9.3 shows a few more regular polygons.



Figure 9.3: Some Regular Polygons

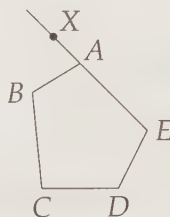
As with quadrilaterals, we'll be focusing on convex polygons in this chapter unless otherwise stated, which means that the interior angles of the polygons are all less than 180° .

Extra! *The most distinct and beautiful statement of any truth must at last take mathematical form.*

—Henry David Thoreau

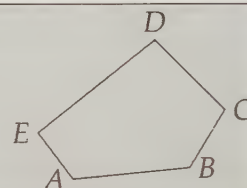
9.2 Angles in a Polygon

In this section, we explore not only the interior angles of a polygon, but also the **exterior angles**. Exterior angles are formed by extending the sides of the polygon. For example, $\angle BAX$ in the figure below is an exterior angle of $ABCDE$.



Problems

Problem 9.1: Find the sum of the measures of the interior angles of pentagon $ABCDE$ shown *without measuring the angles*. (Hint: See how we did Problem 8.1.)



Problem 9.2: In this problem we find a formula for the sum of the interior angles of a convex polygon with n sides. (By convex, we mean that all interior angles have measure less than 180° .)

- Draw all the diagonals from a single vertex of a polygon with n sides. In terms of n , how many triangles are formed?
- Use your dissection from part (a) to find a formula for the sum of the measures of the interior angles of a polygon.

Problem 9.3: The diagram shows the 10 exterior angles of a decagon. Find the sum of these angles. (Note: You cannot assume all the exterior angles have the same measure!)

Hints: 517



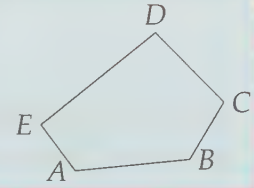
Problem 9.4: Find the number of sides in a regular polygon in which each interior angle measures 172° .

Problem 9.5: $ABCDEFGHIJKLMNO$ is a regular 15-gon.

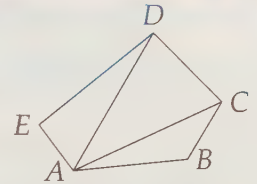
- Find $\angle ACB$.
- Find $\angle ACD$. **Hints:** 48
- Find $\angle ADE$. **Hints:** 250

We found the sum of the interior angles of a quadrilateral by using a diagonal to cut the quadrilateral into triangles. This strategy works for all polygons.

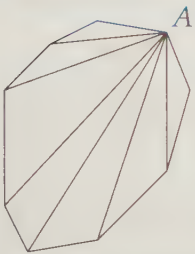
Problem 9.1: Find the sum of the measures of the interior angles of pentagon $ABCDE$ shown *without measuring the angles*.



Solution for Problem 9.1: We don't know anything about pentagons, but we know plenty about triangles. Therefore, we use diagonals to cut the pentagon into triangles. The sum of the angles in the pentagon is the same as the sum of the angles in the three triangles $\triangle ABC$, $\triangle ACD$, and $\triangle ADE$. Each triangle contributes 180° to the total, so the sum of the interior angles in a pentagon is $3(180^\circ) = 540^\circ$.



Problem 9.2: Find a formula for the sum of the interior angles of a convex polygon with n sides.



Solution for Problem 9.2: As we did with the pentagon, we can break any polygon into triangles by drawing all the diagonals from one of the vertices. The sum of the angles in these triangles equals the sum of the angles in the polygon. Therefore, all we have to do is figure out how many triangles we form when we draw the diagonals. We can use our pentagon example for guidance. Let A be the common vertex of all our triangles. Each side of the polygon, except the two sides connected to A , is the side opposite A in exactly one of our triangles. So, there must be $n - 2$ triangles formed. Since there are 180 degrees in each triangle the sum of the angles in an n -gon is $180(n - 2)$ degrees. (It's a little more complicated to prove, but this result also holds for concave polygons.) $\square$

Important: The sum of the interior angles in an n -sided polygon is $180(n - 2)$ degrees. Therefore, the measure of each of the n interior angles in a regular n -gon is $180(n - 2)/n$ degrees.



Now we turn our attention from the interior angles to the exterior angles.

Problem 9.3: The diagram shows the 10 exterior angles of a decagon. Find the sum of these angles. (Note: You cannot assume all the exterior angles have the same measure!)



Solution for Problem 9.3: Each of the exterior angles is supplementary to one of the interior angles, so we know that

$$\text{Sum of interior angles} + \text{Sum of exterior angles} = 10(180^\circ) = 1800^\circ,$$

since together, the interior and exterior angles make up 10 straight angles. Since the sum of the interior angles of a decagon is $(180^\circ)(10 - 2) = 1440^\circ$, we have

$$\text{Sum of exterior angles} = 1800^\circ - 1440^\circ = 360^\circ.$$

□

In the Exercises at the end of the section, you'll be asked to work through this problem for a general n -gon. You'll thereby prove:

Important:



The sum of the exterior angles in a convex polygon with n sides is 360° . Therefore, the measure of each exterior angle in a regular n -gon is $360^\circ/n$.

Let's try using our new knowledge about the angles of polygons on a couple problems.

Problem 9.4: Find the number of sides in a regular polygon in which each interior angle measures 172° .

Solution for Problem 9.4: We let the polygon have n sides. There are two general approaches we can take to this problem:

Focus on the interior angles. Since the measure of one angle is 172° , and the polygon is regular, the sum of all angles is $172n$ degrees. Since we know the sum is also given by $180(n - 2)$ degrees, we can solve the equation $172n = 180(n - 2)$ to find $8n = 360$, so $n = 45$.

Focus on the exterior angles. Since the measure of each interior angle is 172° , each exterior angle is $180^\circ - 172^\circ = 8^\circ$. The sum of the exterior angles is 360° , so there must be $360/8 = 45$ of them. □

Concept:



Sometimes thinking about the exterior angles of a polygon offers a simpler approach than thinking about the interior angles!

Problem 9.5: $ABCDEFGHIJKLMNO$ is a regular 15-gon. Find $\angle ACB$, $\angle ACD$, and $\angle ADE$.

Solution for Problem 9.5: We start with a diagram, then we find whatever angles we can determine. Since the polygon is a regular 15-gon, each exterior angle is $360/15 = 24$ degrees. Therefore, each interior angle is $180^\circ - 24^\circ = 156^\circ$. Since $\triangle ABC$ is isosceles, $\angle BAC = \angle ACB$. The sum of the angles of $\triangle ABC$ gives us $\angle ABC + \angle BAC + \angle ACB = 180^\circ$, so

$$156^\circ + \angle ACB + \angle ACB = 180^\circ.$$

Therefore, $2\angle ACB = 24^\circ$, so $\angle ACB = 12^\circ$. We also have

$$\angle ACD = \angle BCD - \angle ACB = 156^\circ - 12^\circ = 144^\circ.$$

Turning to $\angle ADE$, we try the same strategy: find $\angle ADC$, then subtract it from $\angle CDE$. We note that $ABCD$ is an isosceles trapezoid in which $\overline{BC} \parallel \overline{AD}$ (make sure you see why), so $\angle ADC = 180^\circ - \angle BCD = 24^\circ$. So, $\angle ADE = 156^\circ - 24^\circ = 132^\circ$. □



Exercises

9.2.1 Complete the table below for regular polygons.

Name	# of Sides	Sum of Interior $\angle$ s	Int. $\angle$ Measure	Ext. $\angle$ Measure
pentagon	5	540°	108°	72°
hexagon	6			
heptagon	7			
octagon	8			
nonagon	9			
decagon	10			
dodecagon	12			
pentadecagon	15			
icosagon	20			
triacontagon	30			

9.2.2 Does the formula for the sum of the interior angles in an n -gon still work if the polygon is concave, as in the figure shown at right?



9.2.3 Each interior angle of a certain regular polygon has measure 160° . How many sides does the polygon have?

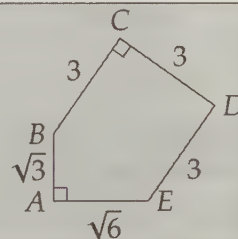
9.2.4 The measures of the angles of a pentagon are in the ratio of $3 : 3 : 3 : 4 : 5$. What is the number of degrees in the measure of the largest angle? (Source: MATHCOUNTS) **Hints:** 587

9.2.5 The sum of the interior angles of a polygon is three times the sum of the exterior angles. How many sides does the polygon have?

9.3 Polygon Area

Problems

Problem 9.6: Find the area of pentagon $ABCDE$ given the side lengths and right angles shown. **Hints:** 538



Problem 9.7: In this problem we find the area of a regular hexagon with side length 8.

- Draw all three long diagonals of the hexagon; what kind of triangles do you form?
- Find the area of the hexagon by finding the areas of the triangles from (a).

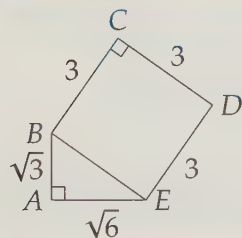
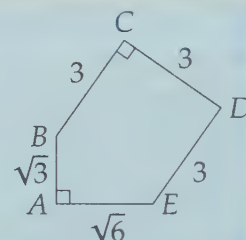
Problem 9.8: In this problem we will find the area of regular octagon $ABCDEFGH$ with side length 4.

- Extend sides $\overline{AB}$, $\overline{CD}$, $\overline{EF}$, and $\overline{GH}$ to meet at points W , X , Y , Z . What type of quadrilateral is $WXYZ$?
- Find the area of $WXYZ$.
- Use $[WXYZ]$ to find the area of $ABCDEFGH$.

Problem 9.9: The center of a regular polygon is the point inside the regular polygon that is equidistant from all of the vertices of the polygon. Let the distance from the center of a regular n -gon to one of its sides be k and the length of each side of the n -gon be s . Break the polygon into triangles by connecting the center of the polygon to each vertex. Find a formula for the area of the n -gon in terms of k , n , and s .

We've seen all sorts of problems involving triangle and quadrilateral areas. You won't be at all surprised that we rely heavily on triangles and quadrilaterals when we find the areas of polygons with more sides.

Problem 9.6: Find the area of pentagon $ABCDE$ given the side lengths and right angles shown.



Solution for Problem 9.6: We start by drawing $\overline{BE}$ to separate right triangle $\triangle ABE$ from quadrilateral $BCDE$. We have $[ABE] = (\sqrt{3})(\sqrt{6})/2 = 3\sqrt{2}/2$, so all we need is $[BCDE]$. From the Pythagorean Theorem, we have $BE = \sqrt{AB^2 + AE^2} = 3$, so $BCDE$ is a rhombus. Since one of its angles is a right angle, it must also be a square. Therefore, $[BCDE] = 9$ and $[ABCDE] = [BCDE] + [ABE] = 9 + 3\sqrt{2}/2$. $\square$

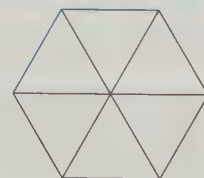
Concept: Break complicated areas into more manageable pieces.



Let's find the areas of a couple regular polygons.

Problem 9.7: Find the area of a regular hexagon with side length 8.

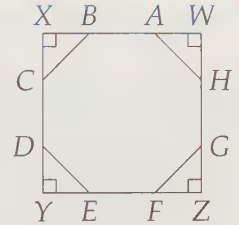
Solution for Problem 9.7: We start by breaking the hexagon into pieces we can handle. Since the angles of a regular hexagon are each 120° , the long diagonals split each angle into two 60° angles. Therefore, each of the little six triangles formed by drawing all three long diagonals is equilateral. Thus, the area of the hexagon is 6 times the area of an equilateral triangle with side length 8. As we saw in Problem 6.11 on page 143, the area of such a triangle is $8^2 \sqrt{3}/4 = 16\sqrt{3}$. So, the area of our regular hexagon is $6(16\sqrt{3}) = 96\sqrt{3}$. $\square$



Concept: Many problems involving regular hexagons can be tackled by dissecting the hexagons into equilateral triangles.

Problem 9.8: Find the area of regular octagon $ABCDEFGH$ with side length 4.

Solution for Problem 9.8: We can find the area with a dissection (try drawing $\overline{AF}$, $\overline{BE}$, $\overline{CH}$, and $\overline{DG}$), but we can find a more elegant approach. We know each angle of a regular octagon is 135° . Since these angles are supplements of 45° angles, we start thinking about 45-45-90 triangles. This makes us think about squares, too. We extend the sides of the octagon, and we form square $WXYZ$. Our octagon is just $WXYZ$ with the four corners (the four 45-45-90 triangles) lopped off.



We can handle squares and 45-45-90 triangles. Since $AH = 4$, we have $WH = 4/\sqrt{2} = 2\sqrt{2}$. Similarly, each of the legs of the little right triangles has length $2\sqrt{2}$. Therefore, the area of each little triangle is $(2\sqrt{2})(2\sqrt{2})/2 = 4$. Since

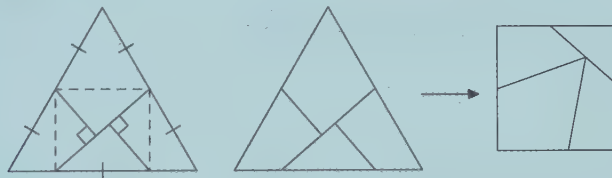
$$[WXYZ] = WZ^2 = (WH + HG + GZ)^2 = (4 + 4\sqrt{2})^2 = 48 + 32\sqrt{2},$$

the area of our octagon is

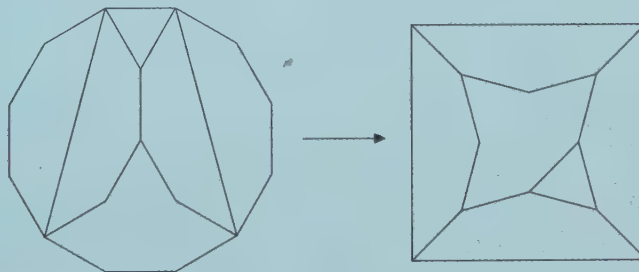
$$[ABCDEFGH] = [WXYZ] - 4[WAH] = 32 + 32\sqrt{2}.$$

□

Sidenote: The figure below shows how an equilateral triangle can be dissected into four pieces that can be reassembled to form a square.



Similarly, a regular dodecagon can be dissected into six pieces that can be reassembled to form a square.



Can you find more regular polygons that can be dissected to form other regular polygons?

We now have another area-finding tool:

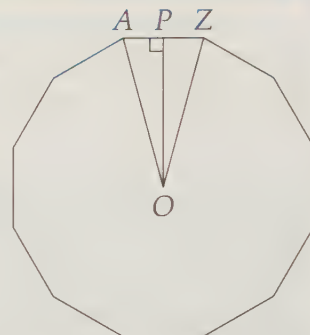
Concept: Sometimes we can view our desired area as the 'leftover' portion from having simple pieces taken away from a simple starting figure.



Now we turn to the general problem of finding the areas of regular polygons.

Problem 9.9: The center of a regular polygon is the point inside the regular polygon that is equidistant from all of the vertices of the polygon. Let the distance from the center of a regular n -gon to one of its sides be k and the length of each side of the n -gon be s . Find a formula for the area of the n -gon in terms of k , n , and s .

Solution for Problem 9.9: Let O be the center of the polygon. (You will be asked to prove that such a point exists as a Challenge Problem.) We can cut the regular polygon into n triangles by connecting the center to every vertex. One such triangle is $\triangle OAZ$, shown in the diagram. The altitude to side AZ of this triangle is the distance from O to a side of the polygon, given in the problem as k . Since AZ is also given as s , we have $[AZO] = (AZ)(OP)/2 = ks/2$. Finally, the polygon consists of n of these triangles, so the area of the whole polygon is $nks/2$. $\square$



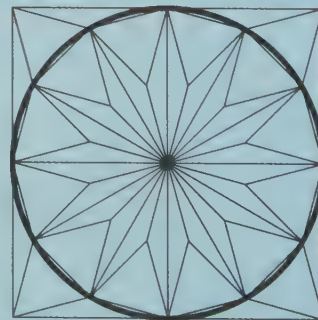
The distance from the center to the sides of a regular polygon is sometimes given the obscure name of the **apothem**. Noting that the perimeter of the polygon is ns , we can write our area formula as:

Important: The area of a regular polygon is half its perimeter times the distance from the center of the polygon to a side.



As we have seen in earlier problems, this isn't always the fastest way to find the area. (Try it on the dodecagon above if you want to torment yourself.)

Sidenote: One particularly fascinating dissection is that of a regular dodecagon shown at right. See if you can figure out how to use the dissection to prove that the area of the dodecagon is 3 times the square of the radius of the dodecagon's circum-circle, which is shown in bold in the diagram. This dissection of a regular dodecagon is used as the logo for the Art of Problem Solving Foundation, which supports problem solving activities for eager middle and high school students. You can find more information on the Foundation (and a graphical proof of the relationship described above) at www.artofproblemsolving.org.



EXERCISES

9.3.1 $ABCDEF$ is a regular hexagon with side length 9.

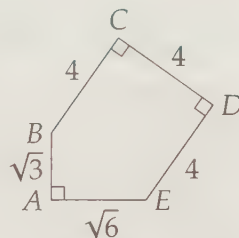
- (a) Find AD .
- (b) Find $[ABCDEF]$.
- (c)★ Find AC .

9.3.2 Find a formula for the area of a regular hexagon with side length s .

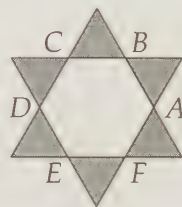
9.3.3 What's wrong with pentagon $ABCDE$ shown at right?

9.3.4 $ABCDEFGH$ is a regular octagon with side length 8. Diagonals $\overline{AE}$ and $\overline{CG}$ meet at X . Point M is the midpoint of $\overline{AB}$.

- (a) Find the area of $ABCDEFGH$.
- (b) Find XM , the apothem of the octagon.
- (c) Find XC . **Hints:** 22



9.3.5 Points A, B, C, D, E , and F are the vertices of a regular hexagon and also trisect the sides of the large equilateral triangles shown. Given that the area of $ABCDEF$ is 24, what is the total area of the shaded regions? (Source: MATHCOUNTS)



9.3.6 We solved Problem 9.8 by extending the sides of a regular octagon to form a square. We didn't, however, prove that we form a square when we connect the points where these extensions meet. Fix this oversight by providing the proof. **Hints:** 564

9.3.7★ In this section we assumed that the long diagonals of a regular hexagon are concurrent. In this problem we fix this oversight by proving that these diagonals are concurrent.

- (a) Let the hexagon be $ABCDEF$ and let point O be the intersection of the bisectors of $\angle A$ and $\angle B$. Prove that $\triangle AOB$ is equilateral.
- (b) Draw $\overline{OC}$. Prove that $\triangle BOC$ is equilateral.
- (c) Prove that $\triangle COD$ is equilateral and that $\overrightarrow{AO}$ goes through D .
- (d) Prove that the long diagonals of $ABCDEF$ all meet at the same point.

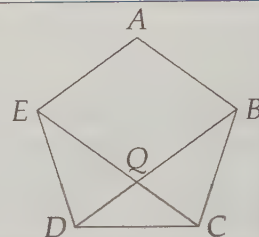
Sidenote: Back on page 251 we challenged you to find pairs of regular polygons such that the first can be dissected and rearranged to form the other. If you didn't find any, keep looking! Not only are there more than the pair we showed you, but in fact any two polygons with the same area can be dissected into the same set of pieces. Therefore, if we have two regular polygons with the same area, we can find some way to dissect one and rearrange the pieces to form the second. Of course, some of these dissections are very hard to find and may involve many pieces!

Many of these dissections are shown at Wolfram Research's MathWorld page; a link to these is provided on the links page described on page viii.

9.4 Polygon Problems

Problems

Problem 9.10: Diagonals $\overline{BD}$ and $\overline{CE}$ of regular pentagon $ABCDE$ meet at Q . Find $\angle BQE$.

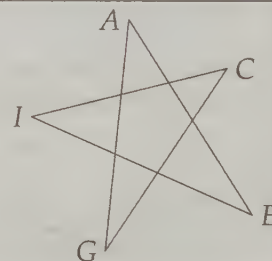


Problem 9.11: In this problem we will find a formula for the number of diagonals in a polygon with n sides.

- How many diagonals does a square have?
- How many diagonals does a pentagon have?
- How many diagonals does a hexagon have?
- Let A be a vertex of a polygon with n sides. How many diagonals of the polygon have A as a vertex?
- Use your answer to the previous part to find a formula for the number of diagonals in a polygon with n sides. Be sure to compare your result to your answers in the first 3 parts to check your formula.

Problem 9.12: $ABCDEFGH$ is a regular octagon. Find the ratio AE/BH . **Hints:** 422

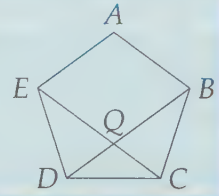
Problem 9.13: In the star diagram, find the sum $\angle A + \angle C + \angle E + \angle G + \angle I$. (You cannot assume the pentagon in the middle of the star is regular!)



Problem 9.14: $ABCDEFGH$ is a regular octagon with side length 12. In this problem we will find the area of $ABDG$.

- What kind of quadrilateral is $ABDG$?
- Consider $FGDE$. Draw altitudes from F and E to $\overline{GD}$ to find GD . (And see if you can find GD in some other clever way!)
- Use your work from the last part as a guide to find the distance from B to $\overline{GD}$.
- Find $[ABDG]$.

Problem 9.10: Diagonals $\overline{BD}$ and $\overline{CE}$ of regular pentagon $ABCDE$ meet at Q . Find $\angle BQE$.



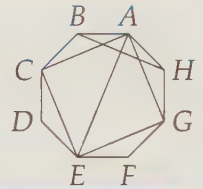
Solution for Problem 9.10: We start by finding the angles we can find. Each interior angle of a regular pentagon is 108° . Since $\triangle EDC$ is isosceles, $\angle DCE = \angle DEC = (180^\circ - 108^\circ)/2 = 36^\circ$. Similarly, we have $\angle CDB = 36^\circ$. Therefore, $\angle CQD = 180^\circ - \angle QDC - \angle QCD = 108^\circ$, so we have $\angle BQE = \angle CQD = 108^\circ$. $\square$

Problem 9.11: Derive a formula for the number of diagonals in a polygon with n sides.

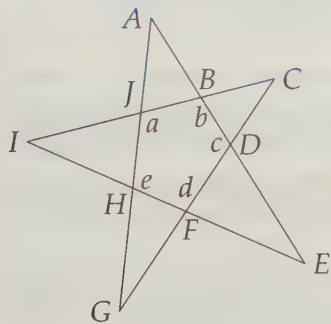
Solution for Problem 9.11: We consider a single vertex, A . A is connected to 2 other vertices by sides, leaving $n - 3$ vertices to which A can be connected by diagonals. Since there are n vertices and each can be connected to $n - 3$ others by diagonals, there appear to be $n(n - 3)$ diagonals. However, this counts each diagonal twice, once for each endpoint. Therefore, we must divide by two to count each diagonal only once. This gives us $n(n - 3)/2$ diagonals in an n -gon. $\square$

Problem 9.12: $ABCDEFGH$ is a regular octagon. Find the ratio AE/BH .

Solution for Problem 9.12: It's not immediately obvious how to compare BH to AE , but there are several other diagonals with the same length as $\overline{BH}$. Considering these, we see that four of them form square $AGEC$. Since $\overline{AE}$ is a diagonal of this square, we have $AE/BH = AE/AG = \sqrt{2}$. $\square$



Problem 9.13: In the star diagram, find the sum $\angle A + \angle C + \angle E + \angle G + \angle I$. (You cannot assume the pentagon in the middle of the star is regular!)



Solution for Problem 9.13: All we have to work with is what we know about the sums of the angles in pentagons and in triangles. We label the angles in the pentagon as shown. We can find the non-star angles of the outer triangles in terms of these (for example, $\angle IJH = 180^\circ - a$). However, if we use a little ingenuity, we don't even need the exterior angles of $BDFHJ$. Consider the angles in each triangle like $\triangle AHE$. These must sum to 180° , so we have:

$$\angle A + e + \angle E = 180^\circ$$

$$\angle C + a + \angle G = 180^\circ$$

$$\angle E + b + \angle I = 180^\circ$$

$$\angle G + c + \angle A = 180^\circ$$

$$\angle I + d + \angle C = 180^\circ$$

We're looking for $\angle A + \angle C + \angle E + \angle G + \angle I$, and we know that $a + b + c + d + e = (180)(5 - 2) = 540^\circ$, since these angles together make up the central pentagon. Therefore, our triangle equations beg to be added together, giving:

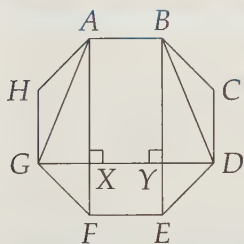
$$2(\angle A + \angle C + \angle E + \angle G + \angle I) + (a + b + c + d + e) = 5(180^\circ).$$

Therefore, $2(\angle A + \angle C + \angle E + \angle G + \angle I) + 540^\circ = 900^\circ$, so

$$\angle A + \angle C + \angle E + \angle G + \angle I = 180^\circ.$$

□

Problem 9.14: $ABCDEFGH$ is a regular octagon with side length 12. Find the area of $ABDG$.



Solution for Problem 9.14: Our $ABDG$ is a trapezoid (you'll prove this explicitly in the Exercises). We'd like to dissect the octagon into pieces we can handle, and we know altitudes of trapezoids can often be useful, so we draw diagonals $\overline{AF}$ and $\overline{BE}$. These are each perpendicular to $\overline{GD}$ and $\overline{FE}$ (you'll prove this fact in the Exercises, too).

Since $\overline{GD} \parallel \overline{FE}$, we have $\angle FGD = 180^\circ - \angle GFE = 45^\circ$, so $\triangle GXF$ is a 45-45-90 triangle. Therefore, $GX = GF/\sqrt{2} = 6\sqrt{2}$. Similarly, $YD = 6\sqrt{2}$. $XYEF$ is a rectangle, so $XY = FE = 12$. Therefore, base $\overline{GD}$ of our trapezoid has length $12 + 12\sqrt{2}$. Before we go through tricky computations to get AX , we can note that by symmetry we have $AF = DG$ (alternatively, we can use HL Congruence to show $\triangle AXG \cong \triangle DXF$), so $AX = AF - XF = DG - GX = 12 + 6\sqrt{2}$. Therefore, our area is:

$$\begin{aligned} [ABDG] &= \frac{(AB + DG)(AX)}{2} = \frac{(24 + 12\sqrt{2})(12 + 6\sqrt{2})}{2} \\ &= \frac{72(2 + \sqrt{2})(2 + \sqrt{2})}{2} = 36(6 + 4\sqrt{2}) = 216 + 144\sqrt{2}. \end{aligned}$$

□

Concept: Most polygon problems are really quadrilateral or triangle problems. When stuck with a polygon, try dissecting it into smaller pieces.



In our last solution, we used **symmetry** to note that $AF = DG$. By symmetry, we essentially mean that the general properties of all the vertices are the same. For example, the distance from one vertex to a vertex three vertices away on the regular polygon (such as AF or DG) is the same no matter which initial vertex we are talking about. However, we couldn't say that ' $AC = BE$ by symmetry' in the last problem, because AC is the distance between two vertices with one vertex between them, while BE is the distance between two vertices with two vertices (C and D) between them.

Concept: Symmetry can be a very useful tool in problems involving regular polygons!



In the Exercises that follow, you'll be asked to solve several problems that could be solved using symmetry. In addition to solutions involving symmetry, try to find explicit solutions using tools such as congruent and similar triangles.

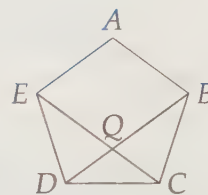
Exercises

9.4.1 $ABCDE$ at right is a regular pentagon. Show that $AEQB$ is a rhombus.

9.4.2 $UVWXYZ$ is a regular hexagon with center O . $UOWABC$ is also a regular hexagon with $AB = 12$.

(a) Find $[UOWABC]$.

(b) Find $[UVWO]$.



9.4.3 All of the interior angles of octagon $EFGHIJKL$ are 135° , $EF = GH = IJ = KL = 2$ and $FG = HI = JK = LE = 6$.

(a) Find the perimeter of the octagon.

(b)★ Find the area of the octagon. **Hints:** 277

(c)★ Find EI . **Hints:** 406

9.4.4 Let $ABCDEF$ be a regular hexagon. Find the ratio of the area of $ABCE$ to the area of the hexagon. (Source: ARML) **Hints:** 74

9.4.5 $ABCDEFGH$ is a regular octagon. Prove that $ACEG$ is a square.

9.4.6 Explicitly prove that $ABDG$ of Problem 9.14 is a trapezoid.

9.4.7 Explicitly prove that $\angle FXG$ is a right angle in Problem 9.14.

9.4.8 The midpoints of the sides of a regular hexagon $ABCDEF$ are joined in order to form a smaller regular hexagon. What fraction of the area of $ABCDEF$ is enclosed by the smaller hexagon? (Source: AMC 12)

9.5 Construction: Regular Polygons

Problems

Problem 9.15: Construct a square.

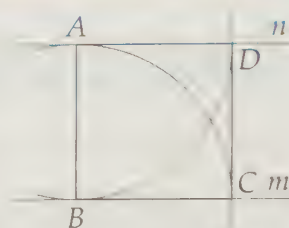
Problem 9.16: Construct a regular hexagon.

Problem 9.17: Construct a regular octagon.

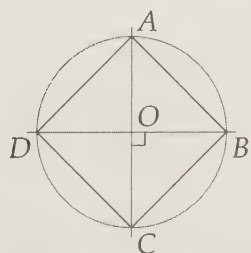
We already know how to construct a regular triangle, so we'll move on to a regular quadrilateral.

Problem 9.15: Construct a square.

Solution for Problem 9.15: Solution 1: A natural solution is to start with one side, $\overline{AB}$, then construct the others. The other two vertices must be on lines through A and B that are perpendicular to $\overline{AB}$, so we construct lines m and n perpendicular to $\overline{AB}$ as shown. We can then find C on m such that $AB = BC$ by constructing a circle with center B and radius AB . Where this circle meets m gives us C . Similarly, we construct a circle with center A and radius AB to find D on n such that $AD = AB$.



We'll leave the proof that this construction does indeed result in $ABCD$ being a square for an Exercise.



Solution 2: We can find an even slicker construction by thinking a little more about squares. For example, we know that the diagonals of a square are perpendicular and they meet at a point that is equidistant from all four vertices. In other words, the intersection of the diagonals of a square is the center of the circumcircle of the square.

We can use this observation to start with a circle to make our square. We draw a circle with center O . We then draw two perpendicular lines through O and label the points where these lines hit the circle A , B , C , and D . Since the four triangles that meet at O are congruent 45-45-90 triangles, we have $AB = BC = CD = DA$. Moreover, each angle of $ABCD$ equals $45^\circ + 45^\circ = 90^\circ$. Therefore, $ABCD$ is a square. $\square$

Let's try using this "circumcircle of a regular polygon" idea on another polygon.

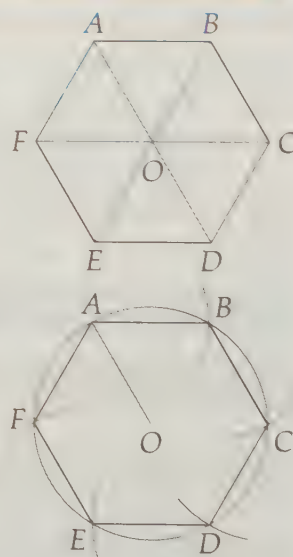
Problem 9.16: Construct a regular hexagon.

Solution for Problem 9.16: Solution 1: We already know how to make an equilateral triangle, and a regular hexagon can be built from six equilateral triangles. Therefore, we can start with equilateral $\triangle AOB$, then construct equilateral $\triangle OBC$ on $\overline{OB}$, then construct equilateral $\triangle OCD$ on $\overline{OC}$, and so on. After making 6 equilateral triangles, we'll have regular hexagon $ABCDEF$.

But that's an awful lot of work. There must be a faster way.

Solution 2: Since the circumcircle worked so well with the square, we'll try it with the hexagon. Since the long diagonal of a regular hexagon is equal both to twice the side length of the hexagon and to the diameter of the circumcircle of the hexagon, the radius of the circumcircle equals the length of a side of the regular hexagon. Therefore, if we draw a circle with radius OA , the vertices of a regular hexagon inscribed in the circle will be at intervals of length OA around the circle.

So, we draw an arc with center A and radius OA to find point B on the circle. Then we draw an arc with center B and radius OA to get C , and so on around the circle. The resulting $ABCDEF$ is a regular hexagon.



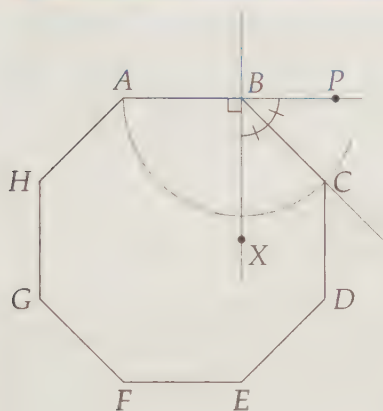
We can prove $ABCDEF$ is regular by noting that each of the triangles formed by connecting consecutive vertices to the center is equilateral. For example, $AO = BO = AB$ means $\triangle AOB$ is equilateral. Similarly, so is $\triangle BOC$, so $\angle ABC = 2(60^\circ) = 120^\circ$. In the same way, we see that each of the angles $ABCDEF$ is 120° , and each of the sides is equal (to the radius of the circle). Therefore, $ABCDEF$ is indeed regular. $\square$

Concept: The circumcircle of a regular polygon can be extremely useful in working with the polygon.



We've already solved a few regular octagon problems, so you should be ready to construct one now.

Problem 9.17: Construct a regular octagon.

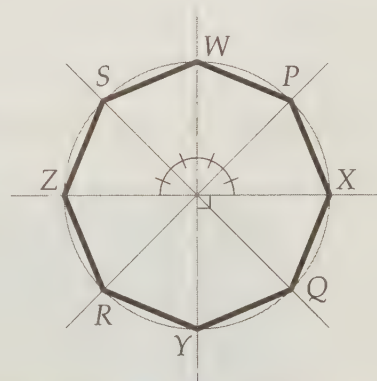


Solution for Problem 9.17: Solution 1: Looking at two consecutive sides of a regular octagon, we see that if we can construct a 135° angle, we can construct a regular octagon. Since $135^\circ = 90^\circ + 45^\circ$ and we know how to make both a 90° and 45° angle, we can make a 135° angle.

Specifically, we start with $\overline{AB}$, which we continue to point P . We then draw a line through B perpendicular to $\overline{AB}$ and let point X be on this line inside the octagon as shown. We bisect $\angle XBP$ and let C be the intersection of this bisector and the circle centered at B with radius BA . Therefore, $AB = BC$ and $\angle ABC = \angle ABX + \angle XBC = 135^\circ$, so C is another vertex of our octagon. We can then copy $\angle ABC$ to make $\angle BCD$, and so on, completing the regular octagon.

But again, that's a lot of work!

Solution 2: Rather than look at the octagon from the point of view of the sides, let's look at it from the center's perspective. Again, we can use the circumcircle. We start by creating square $WXYZ$ by constructing perpendicular lines that meet at the center of the circle. Where these lines meet the circle give us the vertices of square $WXYZ$. The other four vertices of our regular octagon must also form a square, with its vertices at the midpoints of the arcs connecting the vertices of $WXYZ$. To construct these points, we just bisect the right angles formed by $\overline{WY}$ and $\overline{XZ}$. Where these bisectors meet the circle give us the remaining vertices of our regular octagon, $WPXQYRZS$. $\square$



Exercises

9.5.1 Prove that our first solution to Problem 9.15 does produce a square.

9.5.2 Prove that our second solution to Problem 9.17 does produce a regular octagon.

9.5.3 Construct a regular dodecagon.

9.6 Summary

Definitions: A **polygon** is a closed planar figure with line segments as boundaries. As with triangles and quadrilaterals, the segments that form the boundaries are the **sides**, which meet at the **vertices** of the polygon. A **diagonal** is a segment that is not a side, but connects two vertices of a polygon. A **regular polygon** is a polygon in which all the angles have the same measure and all the sides have the same length.

Important:



- The area of a regular polygon is half its perimeter times the distance from the center of the polygon to a side.
- In a polygon with n sides, the measures of the interior angles have a sum of $(n - 2)(180^\circ)$ and the measures of the exterior angles have a sum of 360° .

The names of common polygons and the interior and exterior angle measures for common regular polygons are below:

Name	# of Sides	Int. $\angle$ Measure	Ext. $\angle$ Measure
triangle	3	60°	120°
quadrilateral	4	90°	90°
pentagon	5	108°	72°
hexagon	6	120°	60°
octagon	8	135°	45°
nonagon	9	140°	40°
decagon	10	144°	36°
dodecagon	12	150°	30°

Problem Solving Strategies

Concepts:



- Sometimes thinking about the exterior angles of a polygon offers a simpler approach than thinking about the interior angles!
- Break complicated areas into pieces you can handle. Sometimes we can view a desired area as the 'leftover' portion from having simple pieces taken away from a simple starting figure.
- Many problems involving regular hexagons can be tackled by dissecting the hexagons into equilateral triangles.
- Symmetry can be a very useful tool in problems involving regular polygons.

Continued on the next page. . .

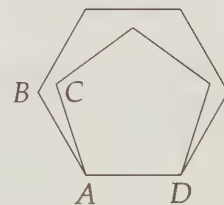
Concepts: . . . continued from the previous page



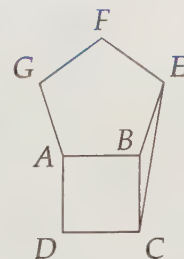
- Most polygon problems are essentially quadrilateral and triangle problems. When stuck with a polygon, try dissecting it into quadrilaterals and triangles.
- The circumcircle of a regular polygon can be extremely useful in working with the polygon.

REVIEW PROBLEMS

- 9.18 How many sides does a regular polygon with interior angles that measure 140° have?
- 9.19 What is the measure of each interior angle of a regular polygon that has 36 sides?
- 9.20 Find the area of a regular hexagon with side length 4.
- 9.21 How many sides does a regular polygon with exterior angles that each measure 6° have?
- 9.22 Find the area of a regular octagon with side length 6.
- 9.23 The measure of each exterior angle of a regular polygon is $1/8$ the measure of an interior angle. How many sides does the polygon have?
- 9.24 $EFGHIJ$ is a regular hexagon. Find EH/EG . **Hints:** 207
- 9.25 A square and a regular hexagon are drawn with the same side length. If the area of the square is $\sqrt{3}$, what is the area of the hexagon? (Source: HMMT)
- 9.26 In a given octagon, four angles have measure x degrees and two have measure $2x$ degrees. The other two angles are right angles. Is the octagon concave or convex?
- 9.27 A given polygon has 27 diagonals. How many sides does the polygon have?
- 9.28 $ABCDEFGH$ is a regular octagon with $AB = 8$. Find AC , AD , and AE .
- 9.29 A regular pentagon and a regular hexagon share a common side $\overline{AD}$, as shown. What is the degree measure of $\angle BAC$? (Source: MATHCOUNTS)
- 9.30 The number of diagonals in a regular polygon is equal to the number of sides. What is the number of degrees in the sum of all the interior angles of the polygon? (Source: MATHCOUNTS)
- 9.31 $ABCDEFGHIJ$ is a regular decagon. Find appropriate congruent triangles (and prove they're congruent) to prove each of the following:
- $AC = AI$.
 - $AD = AH$.
 - $AE = CG$.



9.32 Let $ABCD$ be a square and let $ABEFG$ be a regular pentagon in the diagram at right. Find $\angle BCE$.



9.33 $ABCDEFGH$ is a regular octagon. Diagonals $\overline{AE}$ and $\overline{CG}$ meet at X . Point M is the midpoint of $\overline{AB}$.

- Find $\angle XAB$.
- Find $\angle MXA$.
- Find $\angle MXD$.

Challenge Problems

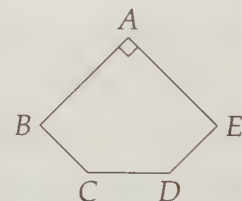
9.34 $A_1A_2A_3 \cdots A_n$ is a regular n -gon.

- Prove that $A_1A_2A_3A_4$ is a trapezoid.
- Prove that $\angle A_2A_1A_4 = 360^\circ/n$. (Source: ARML)

9.35 In our discussion of regular polygons, we assumed that every regular polygon has a point that is equidistant from all the vertices. We called this point the 'center' of the polygon.

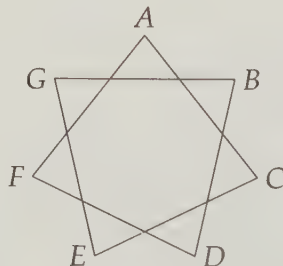
- Prove that inside every regular polygon $\mathcal{P}$ there is a point O that is equidistant from all of the vertices of $\mathcal{P}$. **Hints:** 120
- Let O be the center of regular polygon $\mathcal{P}$, and A be one of the vertices of $\mathcal{P}$. Prove that $\overline{OA}$ bisects an interior angle of $\mathcal{P}$.

9.36 The pentagon $ABCDE$ at right has a right angle at A , $AB = AE$, and $ED = DC = CB = 1$. If $BE = 2$ and $\overline{BE} \parallel \overline{CD}$, what is the area of the pentagon? **Hints:** 191



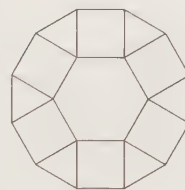
9.37 The total number of degrees in the sum of the interior angles of two regular polygons is 1980. The sum of the number of diagonals in the two polygons is 34. What is the positive difference between the numbers of sides of the two polygons? (Source: MATHCOUNTS) **Hints:** 245

9.38 Find the sum of the angles at the points of the 7-pointed star $ABCDEFG$ shown. Do not assume that the heptagon in the center is regular. **Hints:** 408, 107



9.39 A regular hexagon and an equilateral triangle have the same perimeter. What is the ratio of the area of the hexagon to the area of the triangle?

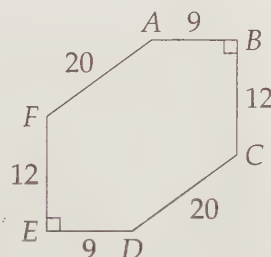
9.40 In a patio, a pattern is determined by regular hexagonal tiles, square tiles and equilateral triangular tiles as shown in the diagram. If the area of each hexagonal tile is 96 in^2 , what is the number of square inches in the area of each square tile? What is the area of the whole tiled region shown? (Source: MATHCOUNTS) **Hints:** 362



9.41 Given that $ABCDEF$ is a regular hexagon with side length 12, find the area of $\triangle ACE$.

9.42 The interior angles of a convex polygon form an arithmetic progression with a common difference of 4° . Determine the number of sides of the polygon if its largest interior angle is 172° . (Source: USAMTS) **Hints:** 123

9.43 Opposite sides of hexagon $ABCDEF$ shown are parallel. Given that $AD = 25$ and the side lengths of the hexagon are as shown, find $[ABCDEF]$. **Hints:** 15



9.44 Determine with proof the number of positive integers n such that a convex regular polygon with n sides has interior angles whose measures, in degrees, are integers. (Source: USAMTS & AHSME) **Hints:** 571

9.45

- What is the largest possible number of interior angles of a convex pentagon that can have measure 90° ? **Hints:** 225
- What is the largest possible number of interior angles of a convex decagon that can have measure 90° ?
- Find a formula in terms of n for the largest number of interior angles of a convex n -gon that can have measure 90° .

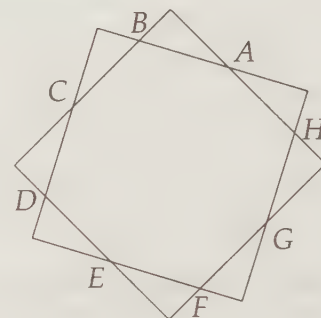
9.46★ It is possible to surround a point with three regular hexagons with side length 1 as shown at left below such that each hexagon shares a side with the other two hexagons. At right below, we see that it is also possible to surround a point with a regular hexagon, a square, and a regular dodecagon such that each has side length 1 and each of the polygons shares a side with the other two.



Find all other groups of three regular polygons with side length 1 that can surround a point such that each polygon shares a side with the other two. Prove that you have found all possible groups of three polygons. **Hints:** 68, 259

9.47★ Let ABC be a right triangle, with $\angle ACB = 90^\circ$. Let P_A , P_B , and P_C be regular pentagons, with side lengths BC , CA , and AB , respectively. Prove that $[P_A] + [P_B] = [P_C]$. **Hints:** 28, 414

9.48★ The two squares shown share the same center O and have sides of length 1. The length of $\overline{AB}$ is $43/99$. Find the area of octagon $ABCDEFGH$ formed by connecting the 8 points where the two squares intersect as shown. (Source: AIME) **Hints:** 56, 442, 233



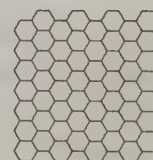
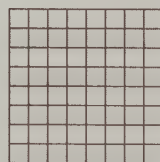
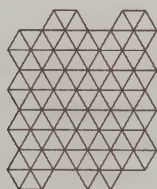
Extra!



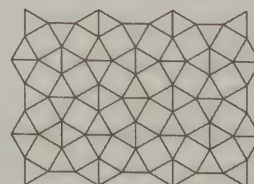
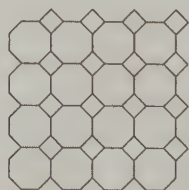
Have you ever seen a bathroom floor tiled with regular pentagons? How about with regular hexagons? Heptagons? Why do squares, triangles, and hexagons seem to show up everywhere, while the other regular polygons don't get much use?

Like lots of other things in geometry, it all boils down to angles. Now that you know how to find the angle in a regular polygon, you can answer this problem for yourself! Take a look at a tiling that does work, such as covering the plane with squares (like a piece of graph paper). At each vertex, exactly four squares meet. The angle of each square is 90° , so the four angles at each vertex add up to 360° .

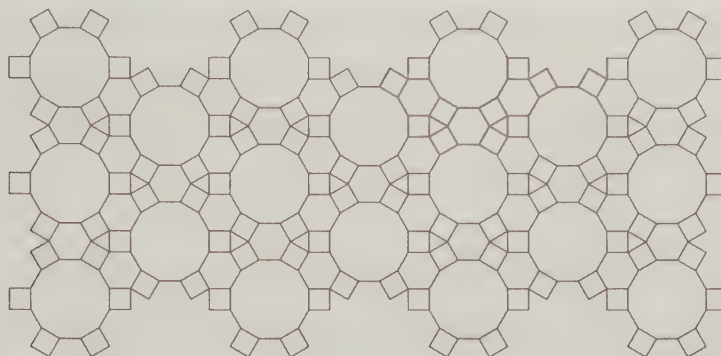
If you solved Problem 9.46, you should be able to quickly identify which regular polygons can be used to tile a plane all by themselves. We call such a regular **tiling** of the plane a **tessellation**. The three tessellations that use only a single regular polygon over and over are shown below. These are called **regular tessellations**.



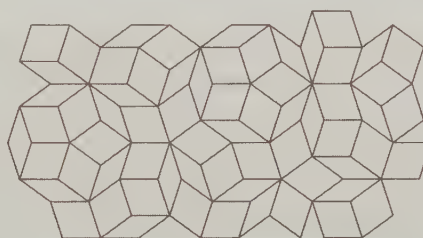
As we saw in Problem 9.46, we have many more possibilities if we allow ourselves to use more than one type of polygon. Shown below are two **semiregular tessellations**, in which the same set of regular polygons surround each vertex in the tessellation. There are eight such tessellations (not including the regular tessellations); see if you can find the other six.



Extra! If we no longer restrict ourselves to having the same polygons around each point, we have **demiregular tessellations**, one of which is shown below.



Of course, we need not only use regular polygons in building tilings. Moreover, tilings need not be periodic (meaning having the same pattern over and over). **Penrose tilings** consist of two quadrilaterals that are used as tiles to make fascinating non-periodic tilings. An example is shown below.



We don't have to restrict ourselves to simple geometric figures to tile a whole plane. The Dutch artist M. C. Escher produced thousands of tessellations using various animal shapes and other whimsical designs. Rather than being satisfied with just a tessellation, Escher often wove his tessellations into very engaging pieces of art. You'll find links to some of Escher's art on the links list described on page viii.

Tessellations have even made their way into puzzles and games, as the two pictures of the tessellation game *Bats and Lizards* below show. Perhaps you can see how these patterns were created from the regular tessellations on the previous page!

